



JEE ADVANCED



Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

Sec: Jr.Super60_NUCLEUS & STERLING BT

2022_P2

Date: 15-10-2023

Time: 09.00Am to 12.00Pm

WTA-23/ WTA-18

Max. Marks:180

MATHEMATICS

: Differentiability at a point, LHD & RHD, Differentiability in an interval & theorems on Differentiability, Problems on continuity and differentiability

PHYSICS

: Entropy, Second law of thermodynamics, Carnot engine, Different types of processes, Efficiency of a cycle, Thermal Properties of Matter : Heat Transfer, Thermal Conduction, Coefficient of Thermal Conductivity, Physical Idea of Flow of Heat At Microscopic Level, : Radiation, Distinction Between Conduction, Convection And Radiation, Good Absorbers and bad reflectors and Vice-Vers, Emittance, Absorptance and Energy Density, Black Body, Pervosty's theory Of Heat Exchange, Kirchoff's Law, Relative Emittance or Emissivity, Newton's Law Of cooling, Radiation Inside A Hallow Enclosure, Stefan's Law, Derivation of Newton's Law of Cooling From Stefan's Law, Distribution of Energy in Black Body Spectrum, Wien's Displacement Law, Solar Constant and Surface Temperature of The Sun, The Green House Effect

CHEMISTRY

: GOC I : Electronic effects: Inductive effect, Inductomeric effect, Electrometric effect, Resonance , mesomeric effect, Aromaticity, Hyperconjugation, Definition of Electrophiles and nucleophiles (Excluding Nucleophilicity comparison)

Name of the Student: _____

H.T. NO:

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JEE-ADVANCE-2022-P2-Model

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+ Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+ 3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+ 4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+ 3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+ Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+ 3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+ 4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+ 3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+ Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Integer Answer Type	+ 3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+ 4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+ 3	-1	4	12
Total				18	60

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MATHEMATICS

Max Marks: 60

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct integer is entered;

Zero Marks: 0 If the question is unanswered;

Negative Marks: -1 In all other cases

- Consider a function $f(x) = \prod_{r=1}^{\infty} \frac{1 + 2\cos\left(\frac{2x}{3^r}\right)}{3}$. Then number of points where $|xf(x)| + ||x-2|-1|$ is non differentiable on interval $(0, 3\pi)$
- Let $f: R \rightarrow R$ be a function defined as $f(x) = \begin{cases} 1-|x|, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$ and $g(x) = f(x-1) + f(x+1) \forall x \in R$. Find number of points of non-differentiability of $g(x)$ on R .
- The total number of points of non-differentiability of $f(x) = \min\left\{|\sin x|, |\cos x|, \frac{1}{4}\right\}$ in $(0, 2\pi)$ is $11p$. Then p is _____
- If n_1 denotes number of points of non-differentiability of $f(x) = \cos(2x-1) \frac{\pi}{2} \cdot |x^2 - 4x - 5| + |x^5 + x^3| + x.e^{|x|}$ and n_2 denotes number of points of discontinuity of $g(x) = \operatorname{sgn}(\sin^2 x - \sin x - 1)$, $x \in (0, 4\pi)$. [Note : $\operatorname{sgn} x$ denotes the signum function of x .] Find the value of $(n_1 + n_2)$.

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$$5. \quad f(x) = \begin{cases} \sin^{-1}\left(\frac{2x}{1+x^2}\right), & x \leq 0 \\ [3x], & 0 < x < k \\ (x-\alpha)^2 + \beta, & x \geq k \end{cases}$$

Where $[p]$ denotes greatest integer less than or equal to p and $k \in \mathbb{N}$. If $f(x)$ is non-differentiable at exactly 7 points, then find the value of $(\alpha + \beta + k)$.

6. If $f(x) = \left|1 - x^3\right| - \left|x^2 - 1\right| + \left|1 - x^2\right| + \left|x^3 - 1\right|$, then the number of points of non-differentiability of $f(x)$ is _____

$$7. \quad \text{If } f(x) = \begin{cases} \frac{\tan[x^2]\pi}{ax^2} + ax^3 + b, & 0 < x \leq 1 \\ 2\cos\pi x + \tan^{-1}x, & 1 < x \leq 2 \end{cases} \text{ is differentiable in } (0, 2], \text{ then } a = \frac{1}{k_1} \text{ and}$$

$$b = \frac{\pi}{4} - \frac{26}{k_2}. \text{ Find } \frac{k_1^2 + k_2^2}{90} \text{ \{where } [.] \text{ denotes greatest integer function}\}.$$

8. If $\lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \cos\frac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $f(x) = e^{-|x|}$ at $x = 0$, then find the value of $(4n + m)$.

SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions
- Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks:** +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

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9. Let $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{px}{n} \sum_{r=1}^n \frac{[r^2 + e^{-x} + r - 1]}{r(r+1)} \right) + \lambda, & x > 0 \\ q, & x = 0 \\ \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\{r^2 + r + e^x - 1\}}{r(r+1)}, & x < 0 \end{cases}$

is differentiable in $\mathbb{R}$: ($[\cdot]$ is G.I.F and $\{ \cdot \}$ is F.P. of x) then.

- A) $p=1$ B) $q=1$ C) $p+q+\lambda=3$ D) $p+q+\lambda=5$

10. Let 'm' is the number of points where $f(x) = \{x\} + \{2x\} + \{3x\}$ is non-differentiable in $[0, n)$ (where $m, n \in \mathbb{N}$ and $\{ \cdot \}$ denotes fractional part function), then which of the following is/are true
- A) If $n=1$, then $m=3$ B) If $n=3$, then $m=11$
 C) If $n=2$, then $m=7$ D) If $n=3$, then $m=12$

11. Let $a, b \in \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$.

Then f is

- A) Differentiable at $x=0$ if $a=0$ and $b=1$
 B) Differentiable at $x=1$ if $a=1$ and $b=0$
 C) Not differentiable at $x=0$ if $a=1$ and $b=0$
 D) Not differentiable at $x=1$ if $a=1$ and $b=1$

12. Given $f(x) = \begin{cases} x^2 \tan^{-1} \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$, then

- A) f is continuous but not derivable at $x=0$
 B) f is differentiable at $x=0$
 C) $f'(x)$ is continuous at $x=0$
 D) $f'(x)$ is not differentiable at $x=0$

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13. Let $f(x) = \begin{cases} (x-1)\sin\left(\frac{1}{x-1}\right), & x \neq 1 \\ 0, & x = 1 \end{cases}$, then

- A) f is neither differentiable at $x = 0$ nor at $x = 1$
 B) f is differentiable at $x = 0$ and at $x = 1$
 C) f is differentiable at $x = 0$ but non-differentiable at $x = 1$
 D) f is differentiable at $x = 1$ but non-differentiable at $x = 0$

14. The function $f(x) = \begin{cases} 5x-4 & \text{for } 0 < x \leq 1 \\ 4x^2-3x & \text{for } 1 < x < 2 \\ 3x+4 & \text{for } x \geq 2 \end{cases}$

then which of the following is **TRUE**

- A) Continuous at $x = 1$ and $x = 2$
 B) Continuous at $x = 1$ but not derivable at $x = 2$
 C) Continuous at $x = 2$ but not derivable at $x = 1$
 D) Continuous both at $x = 1$ and $x = 2$ but not derivable at $x = 1$

SECTION – III (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

15. Let $f(x) = \text{minimum} \left(\cos x, \frac{1}{2}, \{ \sin x \} \right), 0 < x < 2\pi$. Let $x = x_i, i = 1, 2, 3, \dots, n$ be the points

where $f(x)$ is non-differentiable if $\sum_{i=1}^n x_i = \frac{\lambda\pi}{6}$ then value of λ is

[Note: $\{k\}$ denotes the fractional part function of k]

- A) 11 B) 13 C) 19 D) 23



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16. The number of points of non-differentiability of the function $f(x) = \max(\sin x, 2x) + [\max(\sin x, 2x)]$ (where $[\cdot]$ denotes greatest integer function) in $(0, 2\pi)$ is
- A) 12 B) 14 C) 17 D) 18
17. $f(x) = \begin{cases} x^2 \left(\frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} \right); & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then
- A) $f(x)$ is discontinuous at $x = 0$
 B) $f(x)$ is continuous but non-differentiable at $x = 0$
 C) $f(x)$ is differentiable at $x = 0$
 D) $f'(0) = 2$
18. If $f(x) = \begin{cases} |1 - 4x^2|, & 0 \leq x < 1 \\ [x^2 - 2x], & 1 \leq x < 2 \end{cases}$, where $[.]$ denotes the greatest integer function, then $f(x)$ is
- A) Differentiable for all x B) Continuous at $x = 1$
 C) Non-differentiable at $x = 1$ D) None of these

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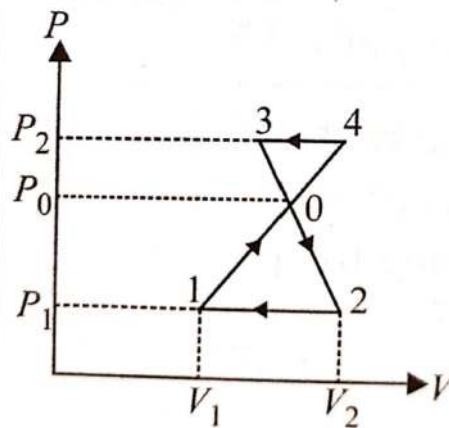
PHYSICS

Max Marks: 60

**SECTION-I
(INTEGER ANSWER TYPE)**

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
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19. In an adiabatic process, the pressure is increased by $\frac{2}{3}\%$. If $\gamma = \frac{3}{2}$, if the decrease in volume is $\frac{n}{9}\%$ then find the value of n .
20. 10 grms of water at 10°C is heated to 50°C . If change in entropy is $7K \ln\left(\frac{323}{283}\right)$, then K is (Specific heat of water is $4.2\text{ J/g}^\circ\text{C}$)
21. Two carnot engines A and B are operated in series. The first one A receives heat at 900 K and rejects to a reservoir at temperature $300n\text{ K}$. The second engine B receives the heat rejected by the first engine and in turn rejects to a heat reservoir at 400 K . When efficiencies of the two engines are equal, find the value of n .
22. The work done by an ideal gas during a closed cycle $1 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$ shown in figure if $P_1 = 10^5\text{ Pa}$, $P_0 = 3 \times 10^5\text{ Pa}$, $P_2 = 4 \times 10^5\text{ Pa}$, $V_2 - V_1 = 10\text{ litre}$, and segments $4 \rightarrow 3$ and $2 \rightarrow 1$ of the cycle are parallel to the V -axis is $250n\text{ J}$. Find n .



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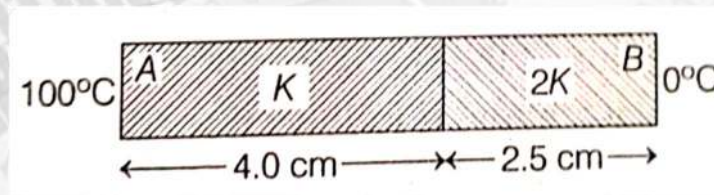
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23. Two spherical bodies A (radius 6 cm) and B (radius 18 cm) are at temperature T_1 and T_2 respectively. The maximum intensity in the emission spectrum of A is at 500 nm and in that of B is at 1500 nm. Considering them to be black bodies, what will be the ratio of the rate of total energy radiated by A to that of B ?
24. A small object is placed at the centre of a large evacuated hollow spherical container. Assume that, the container is maintained at 0K. At time $t = 0$, the temperature of the object is 200 K. The temperature of the object becomes 100 K at $t = t_1$ and 50 K at $t = t_2$. Assume the object and the container to ideal black bodies. The heat capacity of the object does not depend on temperature. The ratio (t_2 / t_1) is.....
25. As per the given figure, two plates A and B of thermal conductivity K and $2K$ are joined together to form a compound plate. The thickness of plates are 4.0 cm and 2.5 cm, respectively and the area of cross-section is 120 cm^2 for each plate. The equivalent thermal conductivity of the compound plate is $\left(1 + \frac{5}{3\alpha}\right)K$, then the value of α will be.....



26. A body cools from 60°C to 40°C in 6 minutes. If temperature of surroundings is 10°C . Then, after the next 6 minutes, its temperature will be $14x$, find x

SECTION – II

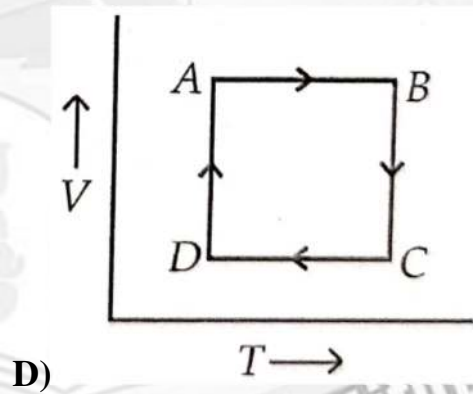
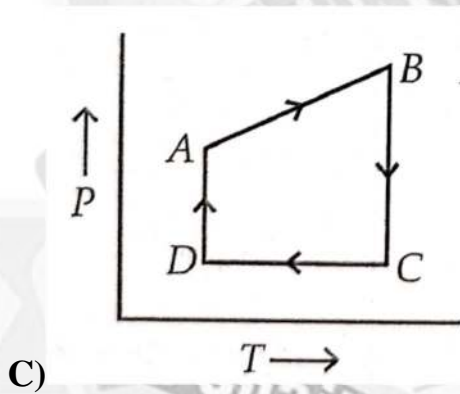
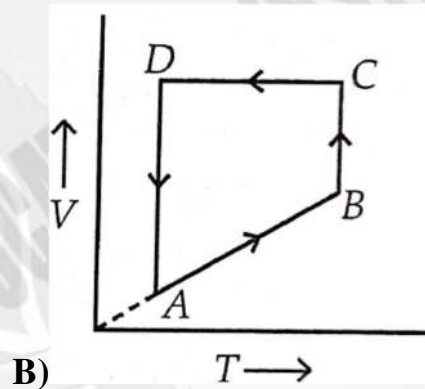
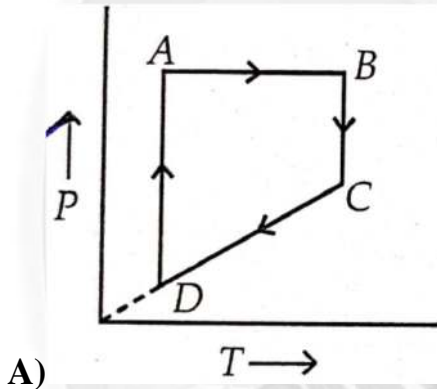
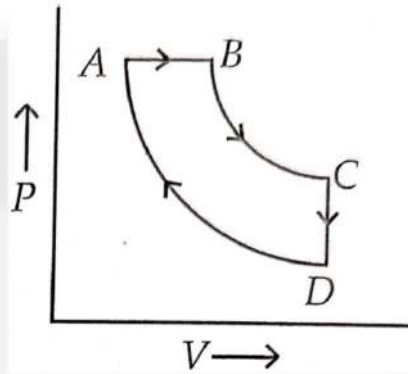
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions.
- Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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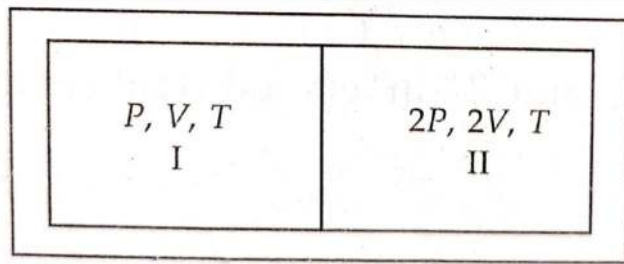


27. A cyclic process $ABCD$ is shown in the $P-V$ diagram. (BC and DA are isothermal)
Which of the following curves represents the same process?

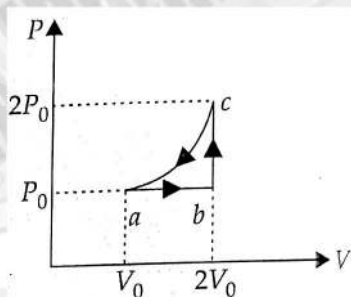


28. A partition divides a container having insulated walls into two compartments I and II. The same gas fills the two compartments whose initial parameters are given. The partition is a conducting wall which can move freely without friction. Which of the following statements is/are correct, with reference to the final equilibrium position?





- A) The pressure in the two compartments are equal
- B) Volume of compartment I is $\frac{3V}{5}$
- C) Volume of compartment II is $\frac{12V}{5}$
- D) Final pressure in compartment I is $\frac{5P}{3}$
29. One mole of an ideal monatomic gas has initial temperature T_0 , is made to go through the cycle $abca$ shown in figure.



If U denotes the internal energy, then choose the correct alternative.

- A) $U_c > U_b > U_a$ B) $U_c - U_b = 3RT_0$ C) $U_c - U_a = \frac{9RT_0}{2}$ D) $U_b - U_a = \frac{3RT_0}{2}$
30. Two bodies A and B have thermal emissivities of 0.01 and 0.81, respectively. The outer surface areas of the two bodies are equal. The two bodies emit total radiant power at the same rate. The wavelength λ_B corresponding to maximum spectral radiance in the radiation from B is shifted from the wavelength corresponding to maximum spectral radiance in the radiation from A , by $1.00 \mu\text{m}$. If the temperature of A is 5802 K then
- A) The temperature of B is 1934 K
- B) $\lambda_B = 1.5 \mu\text{m}$
- C) The temperature of B is 1160 K
- D) The temperature of B is 2901 K

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31. The emissive power of a black body at $T = 300\text{ K}$ is 100 Watt/m^2 . Consider a body B of area $A = 10\text{ m}^2$, coefficient of reflectivity $r = 0.3$ and coefficient of transmission $t = 0.5$. Its temperature is 300 K . Then which of the following are correct?
- A) The emissive power of B is 20 W / m^2
 B) The emissive power of B is 200 W / m^2
 C) The power emitted by B is 200 Watts
 D) The emissivity of B is 0.2
32. A heated body emits radiations which has maximum intensity at frequency ν_m . If the temperature of the body is double, then
- A) the maximum intensity radiation will be at frequency $2\nu_m$.
 B) the maximum intensity radiation will be at frequency $(1/2)\nu_m$.
 C) the total emitted energy will increase by a factor of 16 .
 D) the total emitted energy will increase by a factor of 2 .

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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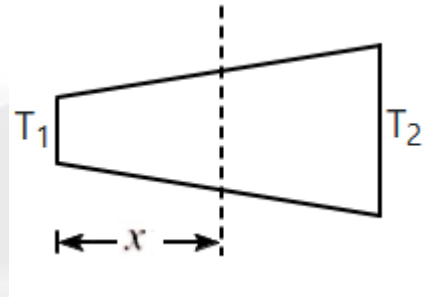
Negative Marks : -1 In all other cases

33. A Carnot engine operating between temperatures T_1 and T_2 has efficiency $1/6$. When T_2 is lowered by 62 K , its efficiency increases to $1/3$. Then T_1 and T_2 are, respectively
- A) 372 K and 310 K B) 372 K and 330 K
 C) 330 K and 268 K D) 310 K and 248 K
34. One mole of oxygen gas is made to undergo a process in which its molar heat capacity C depends on its absolute temperature T as $C = \alpha T$. Work done by it when heated from an initial temperature T_0 to a final temperature $2T_0$, will be
- A) $4\alpha T_0^2$ B) $(\alpha T_0 - 1)\frac{3T_0}{2}$ C) $(3\alpha T_0 - 5)\frac{T_0}{2}$ D) None of these

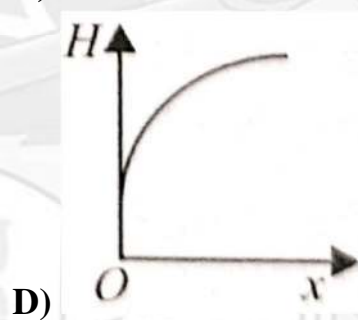
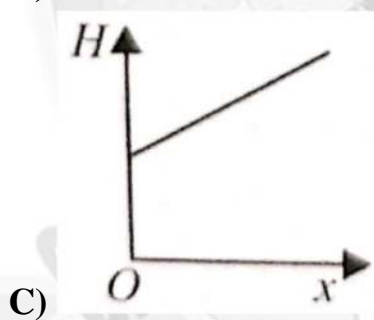
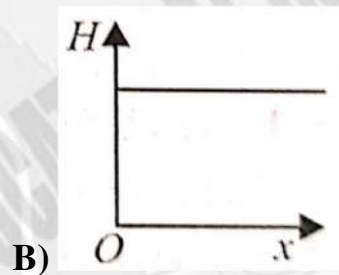
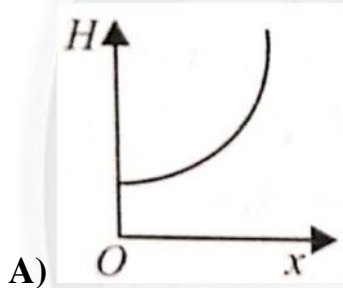
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6th-12th Class
300
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RANKRANK
1JEE Advanced
2023VAVILALA
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341
360
RANKRANK
1NEET
2023BORA VARUN
CHAKRIVARTHI
AIR-1000000
6th-12th Class
720
720
RANKRANK
1



35. Radius of a conductor increases uniformly from left end to right end as shown in figure.



Material of the conductor is isotropic and its curved surface is thermally insulated from surrounding. Its ends are maintained at temperatures T_1 and T_2 ($T_1 > T_2$). If, in steady state, heat flow rate is equal to H , then which of the following graphs is correct?



36. A system S receives heat continuously from an electrical heater of power $10W$. The temperature of S becomes constant at $50^\circ C$ when the surrounding temperature is $20^\circ C$. After the heater is switched off, S cools from $35.1^\circ C$ to $34.9^\circ C$ in 1 minute. The heat capacity of S is

- A) $100 J/^\circ C$ B) $300 J/^\circ C$ C) $750 J/^\circ C$ D) $1500 J/^\circ C$



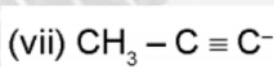
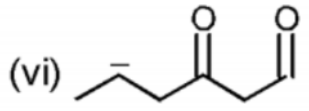
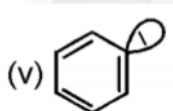
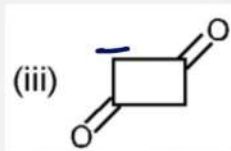
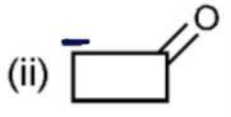
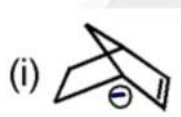
CHEMISTRY

Max Marks: 60

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

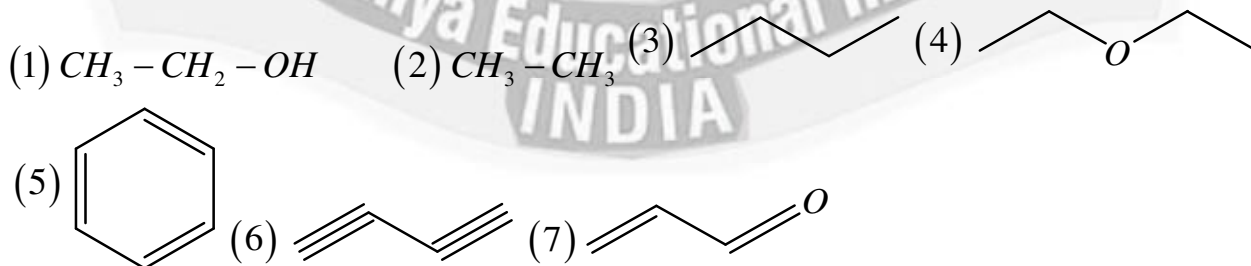
37. In how many species the -ve charge (anion) which is not delocalized by resonance, are :



38. Which of the following statements is (are) true about resonance.

- Resonance is an intramolecular process.
- Resonance involves delocalization of p electrons only
- Resonance decreases potential energy of a molecule
- Resonance is not the only way to increase molecular stability
- The canonical structure explains all features of a molecule
- The resonance hybrid explains all features of a molecule
- Resonance hybrid is real and resonating structures are imaginary
- Resonance hybrid is always more stable than all canonical structures

39. How many can show inductive effect



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341
360
RANK



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CHAKRAVARTHI
AIR-1000000
6th-12th Class
720
720
RANK



**RANK
1**



40. From the following compounds/ions :

- (A) $\overset{\oplus}{C}H_3$ (B) BF_3 (C) NH_3 (D) $NH_2 - NH_2$ (E) $AlCl_3$
 (F) BCl_3 (G) $NaNH_2$

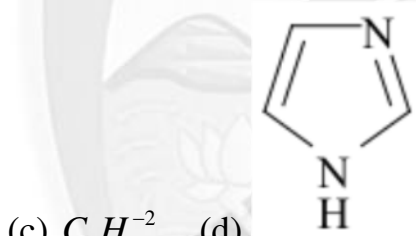
Identify value of "X".

Where "X" is the total number of E^+ / electrophiles.

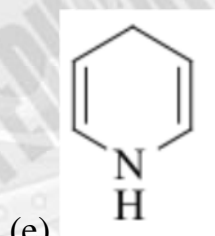
41. From the following compound



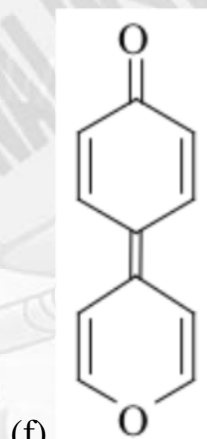
(b) $C_{14}H_{14}$ (use 2 six membered ring)



(d)



(f)

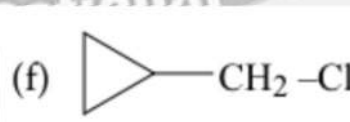
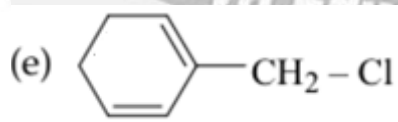
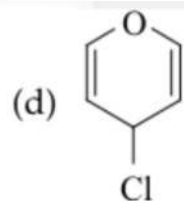
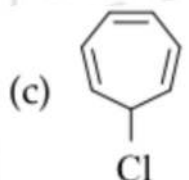


(h)

(i) $C_4H_4^{-2}$

Identify the total number of aromatic compounds.

42. How many products will be aromatic ion when $AgNO_3$ reacts with the following compounds?



(h) $H-C \equiv C-Cl$

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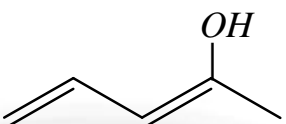
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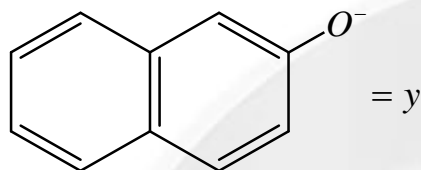
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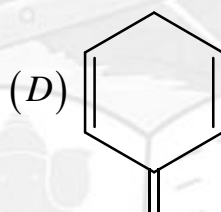
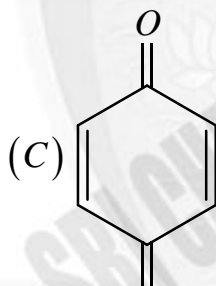
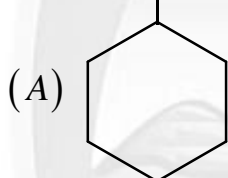
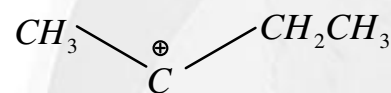


43. Find no. of resonating structure of  = x



then $\frac{x+y}{2}$ is

44. The total number of contributing structures in A, B, C, D showing hyperconjugation (involving C-H bonds) then find the total number of A + B + C + D is



SECTION – II

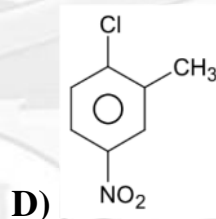
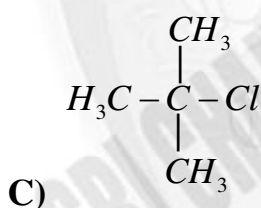
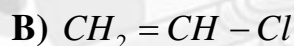
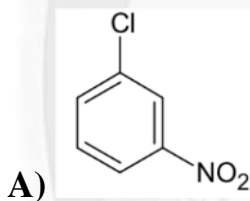
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions.
 - Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
 - For each question, choose the option(s) corresponding to (all) the correct answer(s).
 - Answer to each question will be evaluated **according to the following marking scheme:**
- Full Marks: +4 If only (all) the correct option(s) is(are) chosen; Partial Marks +3 If all the four options are correct but ONLY three options are chosen;
 Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
 Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
 Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
 Negative Marks: -2 In all other cases.

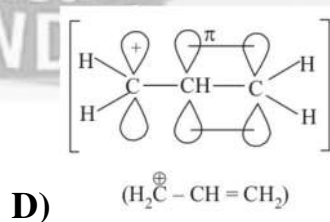
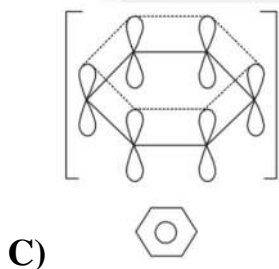
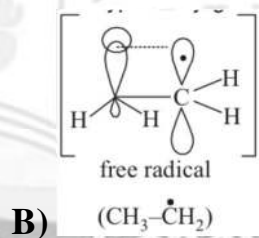
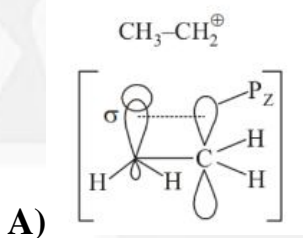




45. Which of the following statement is/are correct?
 A) The energy of resonance hybrid is always less than that of any resonating structure.
 B) The resonance energy is the difference between the enthalpies of formation of the molecule and the resonating structure having maximum energy.
 C) The resonance structures are hypothetical structure and they do not represent any real molecule.
 D) In delocalized structure of benzene the π -charge cloud is spread equally above below the plane of molecule.
46. Which of the following is/are acceptable resonating structures of Buta-1, 2, 3-triene.
 A) $\overset{\ominus}{\text{CH}_2} - \text{C} \equiv \text{C} - \overset{\oplus}{\text{CH}_2}$ B) $\text{CH}_2 = \text{C} = \text{C} = \text{CH}_2$
 C) $\overset{\oplus}{\text{CH}_2} - \text{C} \equiv \text{C} - \overset{\ominus}{\text{CH}_2}$ D) $\dot{\text{C}}\text{H}_2 - \text{C} \equiv \text{C} - \dot{\text{C}}\text{H}_2$
47. In which of the following compound(s) the $\text{C}-\text{Cl}$ bond is shorter than that in $\text{C}_2\text{H}_5\text{Cl}$?

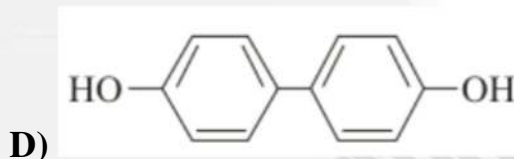
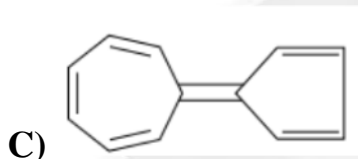
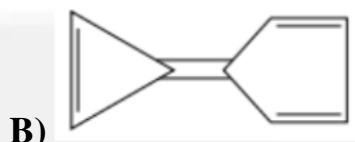
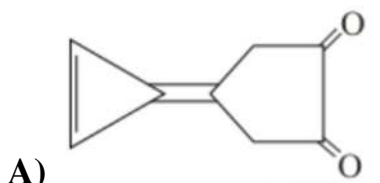


48. Among the following orbital interactions which represents hyper-conjugation interaction?

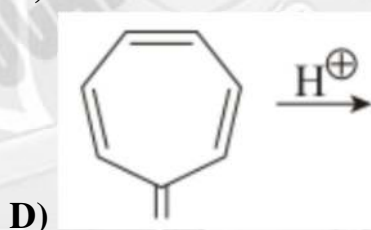
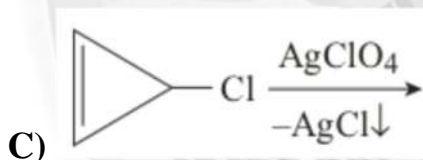
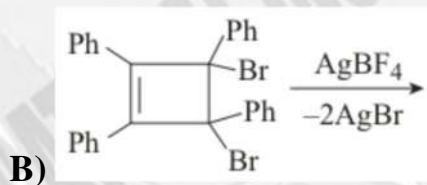
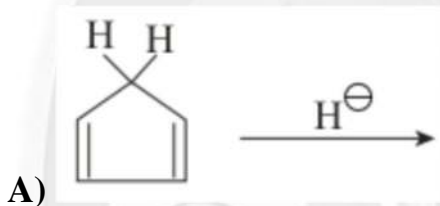




49. Which has non-zero dipole moment?



50. Which of the following reactions give an aromatic product?



SECTION – III (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

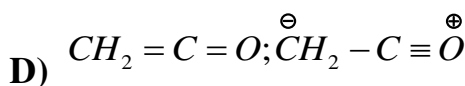
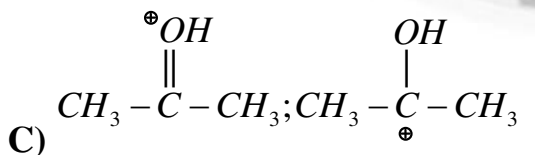
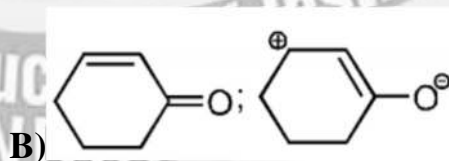
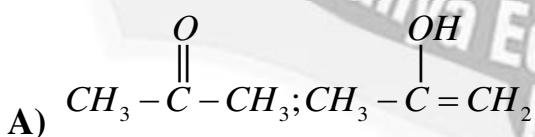
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

51. Which of the following are not resonating structures of each other?



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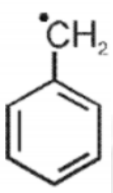


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6th-12th Class
**720
720**
RANK



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1**

52. **Column-I** **Column-II**
- A) $\overset{\oplus}{CH_2} - CH = CH_2$ p) Resonance possible
- B) $H_2N - CH = CH_2$ q) Even number of p-electrons
- C)  r) Localized lone pair of e^-
- D) $H_2N - \overset{\overset{||}{NH}}{C} - NH_2$ s) Delocalized lone pair of e^-
- t) $2e^-$ in p orbitals.
- A) A-pq; B-qs; C-r; D-pqr B) A-pqs; B-qr; C-pq; D-q
- C) A-pqt; B-pqs; C-p; D-pqrs D) A-p; B-qrt; C-pq; D-pqr
53. The stability order of :
- (1) $CH_3 - \overset{\overset{\oplus}{O}}{C} = \overset{\ominus}{O} - H$ (2) $CH_3 - \overset{\overset{||}{O}}{C} - OH$ (3) $CH_3 - \overset{\overset{\oplus}{C}}{C} - OH$
- A) $1 > 2 > 3$ B) $2 > 3 > 1$ C) $2 > 1 > 3$ D) $1 > 3 > 2$
54. **Column-I** **Column-II**
- A) $-\overset{\oplus}{N}Me_3$ p) $+I$
- B) $-\overset{\oplus}{P}Me_3$ q) $-I$
- C) $-OMe$ r) $+R$
- D) $-CH_3$ s) $-R$
- t) $+H$ (hyper conjugation).
- A) A-q; B-qs; C-qr; D-pt B) A-q; B-q; C-r; D-t
- C) A-q; B-s; C-t; D-s D) A-q; B-r; C-t; D-s



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Category Ranks

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RANKS (50%)

BELOW 100
All India Open
Category Ranks

32

BELOW 1000
All India Open
Category Ranks

181

BELOW 100
All India
Category Ranks
Count

89

BELOW 1000
All India
Category Ranks
Count

699

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